

Basic Bash Scripting Commands - Interview Guide

1. Script Structure

- `#!/bin/bash`
- Every Bash script starts with a shebang line to tell the system to use bash.

2. Print and Comments

- `echo 'Hello World'` - Print text to the screen
- `# This is a comment` - Comments start with `#`

3. Variables

- `name='Sai'` - Assign variable
- `echo $name` - Display variable value
- `read user` - Take input from user

4. Conditionals

- `if [ $num -gt 10 ]; then echo 'Greater'; else echo 'Smaller'; fi`

5. Loops

- `for i in 1 2 3; do echo $i; done`
- `count=1; while [ $count -le 5 ]; do echo $count; ((count++)); done`

6. Case Statement

- `case $option in start) echo 'Starting';; stop) echo 'Stopping';; *) echo 'Invalid';; esac`

7. Functions

- `greet() { echo 'Hello $1'; }`
- `greet Sai`

8. Command Substitution

- `today=$(date)`
- `echo 'Today is $today'`

9. File Tests

- if [-f /etc/passwd]; then echo 'File exists'; fi

10. Arithmetic

- a=5; b=3; echo \$((a+b)) - Prints 8

11. Arrays

- fruits=('apple' 'banana' 'mango')
- echo \${fruits[0]} - Prints first item

12. Sleep and Wait

- sleep 5 - Pauses script for 5 seconds

13. Exit Status

- ls /tmp; echo \$? - Displays success/fail status of last command

14. Script Arguments

- echo \$0 - Script name
- echo \$1 - First argument
- echo \$# - Argument count

15. File Operations

- touch new.txt - Create file
- cp file1 file2 - Copy file
- mv file1 /tmp/ - Move file
- rm -f file.txt - Delete file

16. Cron Scheduling

- crontab -e - Edit cron jobs
- 0 17 * * * /home/user/backup.sh - Run every day at 5PM